#B Use HKDF-Expand but return the raw bytes rather than a Key object

Listing 13.10 shows an example of deriving the sender and recipient keys and Common IV based on the test case from Appendix C of the OSCORE specification (https://tools.ietf.org/html/rfc8613#appendix-C.1.1). You can run the code to verify that you get the same answers as the RFC. You can use `org.apache.commons.codec.binary.Hex` to print the keys and IV in hexadecimal to check the test outputs.

> **WARNING** Don't use this master key and master salt in a real application! Fresh keys should be generated for each device.

Listing 13.10 Deriving OSCORE keys and IV

```
public static void main(String... args) throws Exception {
    var algorithm = AlgorithmID.AES_CCM_16_64_128;    #A
    var masterKey = new byte[] {    #B
            0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07,
0x08,    #B
            0x09, 0x0a, 0x0b, 0x0c, 0x0d, 0x0e, 0x0f, 0x10
#B
    };    #B
    var masterSalt = new byte[] {    #B
            (byte) 0x9e, 0x7c, (byte) 0xa9, 0x22, 0x23,
0x78,    #B
            0x63, 0x40    #B
    };    #B
    var hkdfKey = HKDF.extract(masterSalt, masterKey);    #C
    var senderId = new byte[0];    #D
    var recipientId = new byte[] { 0x01 };    #D
```